

Department Overview

- **Department Name:** Department of Mechanical Engineering
- **Faculty:** Faculty of Engineering & Architecture
- **Accreditation:** Fully accredited by the **Pakistan Engineering Council (PEC)**.
- **Core Technological Focus:** Providing engineering solutions for product and process development, ranging from micro-component design to massive industrial plants, machinery, and vehicles. Key areas include transportation, healthcare infrastructure, construction, thermal sensors/devices, and robotics.
- **Career Sectors:** Graduates possess highly diverse skills, securing career opportunities across sectors such as manufacturing, automotive, power generation, construction, biomedical, aerospace, process industries, resource management, and oil & gas.

Vision and Mission

- **Vision:** To be a highly reputed Mechanical Engineering faculty, that offers its own unique blend of technical specializations and research opportunities.
- **Mission:** To provide a modern academically stimulating environment that produces socially responsible young professionals who can solve mechanical engineering challenges and serve ethically.

1. Bachelor of Science in Mechanical Engineering (BS Mechanical)

Admission Requirements

- **Academic Background:** F.Sc (Pre-Engineering) or ICS with Mathematics & Physics from a recognized board (or equivalent qualification) with at least **60% marks**. Alternatively, a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of **60% marks**.

Degree Requirements

- **Total Credit Hours:** 136
- **Total Courses:** 57
- **Minimum Academic Standing:** CGPA $\geq$ 2.0

Program Educational Objectives (PEOs)

Graduates are expected to exhibit the following attributes 3 to 5 years post-graduation:

1. Provide solutions to the problems associated with design, manufacturing, and energy systems using fundamentals of mechanical engineering.
2. Design and investigate mechanical systems by using analytical and modern techniques for innovation, technological development, and sustainability.
3. Acquire the competency to address real-world problems in socio-technological areas through ethical values and professional skills.

Curriculum Plan (Third Board of Studies Schema)

Note: Credit hour listings denote (Theory - Lab).

Semester	Course Code	Course Title	Prerequisite	Credit Hours
1st	CHE-102	Applied Chemistry	None	2-0
	PHY-205 (+L)	Applied Physics	None	2-1
	MATHP-118	Calculus & Analytical Geometry	None	3-0
	HUM-164	Fundamentals of English Grammar	None	2-0
	MECH-201L	Mechanical Engineering Drawing	None	0-2
	MECH-101L / CS-111	Workshop Practice Lab / Intro to Computer Programming	None / None	0-2 / 2-1
2nd	EE-106(+L)	Basic Electronics	None	2-1
	MECH-244	Engineering Materials &	None	3-0

		Metallurgy		
	MECH-121	Thermodynamics - I	None	3-0
	HUM-101 / MECH-141	Islamic Studies / Engineering Mechanics (Statics)	None / None	2-0 / 3-0
	MECH-247L / HUM-268	Computer- Aided Grid Drawing (LAB) / Comm. Skills	None / None	0-1 / 3-0
3rd	MATHA-214	Differential Equations	Calculus & Geometry	3-0
	MECH-241 / 205L	Engineering Mechanics (Dynamics) & Lab	None	3-0 / 0-1
	MECH-222 / 242	Fluid Mechanics - I / Solid Mechanics - I	None / None	3-0 / 3-0
	HUM-102 / EE- 108L	Pakistan Studies / Electrical Engineering	None / None	2-0 / 2-1
4th	MATHA-221	Linear Algebra & Transforms	Calculus & Geometry	3-0

	MECH-223 / 223L	Fluid Mechanics II & Lab	Fluid Mechanics - I	3-0 / 0-1
	MECH-262 / ECON-105	Machine Design - I / Engineering Economics	None / None	3-0 / 2-0
	MECH-221 / 221L	Thermodynami cs-II & Lab	Thermodynami cs - I	3-0 / 0-1
	HUM-276	Technical RPT WRT & PT Skills	None	2-0
5th	MECH-243 / 243L	Solid Mechanics - II & Lab	Solid Mechanics - I	3-0 / 0-1
	MECH-363L	Computer- Aided Engineering Lab	None	0-1
	MECH-364	Mechanical Instrumentatio n & Control	None	2-1
	MATHA-216	Numerical Analysis	Linear Algebra & Transforms	2-1
	MECH-304 / 305	Machine Design - II / Manufacturing	None / None	3-0 / 3-0

		Processes - I		
6th	MECH-306 / 306L	Manufacturing Processes - II & Lab	None	3-0 / 0-1
	STAT-203	Applied Statistics	None	3-0
	MECH-321 / 321L	Heat & Mass Transfer & Lab	None	3-0 / 0-1
	MECH-421	Internal Combustion Engines	None	2-0
	MECH-362 / 362L	Mechanics of Machines & Lab	None	3-0 / 0-1
7th	ENGG-203	Maintenance Engineering	None	2-0
	MECH-322 / 322L	Refrigeration & Air Conditioning & Lab	None	3-0 / 0-1
	ENVIRON-344	Health, Safety & Environment	None	1-0
	MECH-ME / MECH-TE	Management Elective / Technical	None / None	3-0 / 3-0

		Elective - I		
	MECH-462	Project – I	None	0-3
8th	MECH-461 / 461L	Mechanical Vibrations & Lab	None	3-0 / 0-1
	HUM-310	Professional Ethics & Entrepreneurship	None	2-0
	MECH-422	Power Plant	None	3-0
	MECH-TE / MECH-463	Technical Elective - II / Project - II	None / None	3-0 / 0-3

2. Master of Science Program (MS)

Admission Requirements

- **Academic Background:** 16 years of education or an equivalent benchmark qualification (such as a 4-year BS or B.E. degree) from an HEC-recognized university with at least **60% marks** under the annual grading setup or a minimum CGPA of **2.5 out of 4.0**.
(Note: The raw webpage text retains a copy-paste placeholder typo erringly naming Civil/Petroleum engineering fields within this admission block).
- **Testing:** A valid **GAT General** score with at least **50% marks**.

Degree Requirements

- **Total Credit Hours:** 30
- **Total Coursework Requirements:** 8 advanced graduate courses.
- **Minimum Academic Standing:** CGPA $\geq$ 2.5